ENGINEERING ASSESSMENT

Engineering Assessment

Four pillars: Foundations, SQL & Visualisation, Big Data, and Governance

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14 October 2026

PILLAR 01

Foundations

Clean source data and build a warehouse that preserves employee history.

TASK	QUESTION ASKED	WHAT I DID
1.1	Clean projects, parse dates, calculate four metrics, standardise status and assign risk.	Loaded 500 projects with Pandas. Added variance, over-budget flag, duration and utilisation; handled missing amounts/zero budgets; mapped status and prioritised High risk.
1.2	Design a star schema with SCD Type 2 employee history and validate its integrity.	Built date, project, employee and vendor dimensions, a bridge and transaction facts. Added surrogate keys, validity periods and one current employee version; checked overlaps and foreign keys.
1.3	Find all employee quality issues, document fixes and report affected rows.	Checked all 1,000 employees for missing fields, email/date errors, salary/experience ranges and status conflicts. Applied documented fixes, logged counts and saved the correction audit.

Evidence: 500 projects • 1,000 employees • 50,000 transactions • 1,826 salary-history records

PILLAR 02

SQL & Data Visualisation

Business questions, transaction ETL, SQL optimisation and the executive dashboard.

TASK	QUESTION ASKED	WHAT I DID
2.1 Q1	Departments spending >90% of budget.	Grouped budgets/costs; calculated spend percentage and over-budget flag.
2.1 Q2	Current managers with >3 active projects.	Joined current employees; counted and filtered strictly more than 3 active projects.
2.1 Q3	Vendors with >5% of spend; risk labels.	Calculated vendor shares/risk flags and applied the strict >5% spend filter.
2.1 Q4	Pending/disputed totals >50,000 AED.	Filtered status/currency; grouped projects and applied the value threshold.
2.1 Q5	Monthly spend, running total and change.	Used a calendar grid, SUM OVER and LAG to compare monthly/category spending.
2.1 Q6	Top 20 largest employee salary increases.	Compared adjacent SCD2 versions; calculated and ranked salary increases.

2.2 | Asked: ingest/enrich 50K transactions and export a timed summary.

Flattened JSON, parsed dates, used validated LEFT joins and added project / current approver context, approval flag, amount_aed and reporting month.

2.3 | Asked: benchmark, rewrite and index the query.

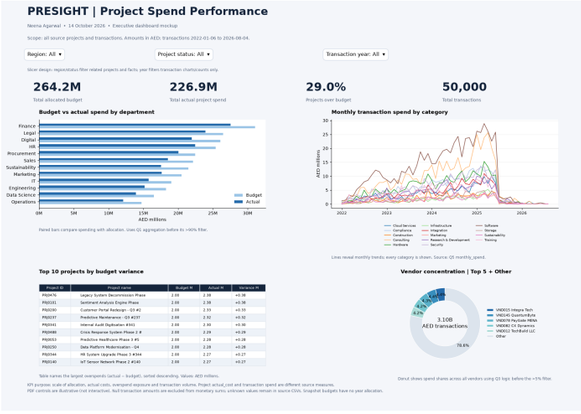
Rewrote joins/filtering, added indexes and verified equivalent result rows. Saved timing evidence: 14.42 ms to 6.14 ms, a measured 2.35x speedup.

2.4 | Asked: an annotated one-page dashboard with six visuals.

Generated the PDF with actual KPIs, budget/spend bars, monthly trends, top-project table, vendor donut and slicer mockups. The dashboard preview is embedded here.

Q3 correctly returns no rows: the largest vendor share is about 4.36%, below 5%.

DASHBOARD PDF

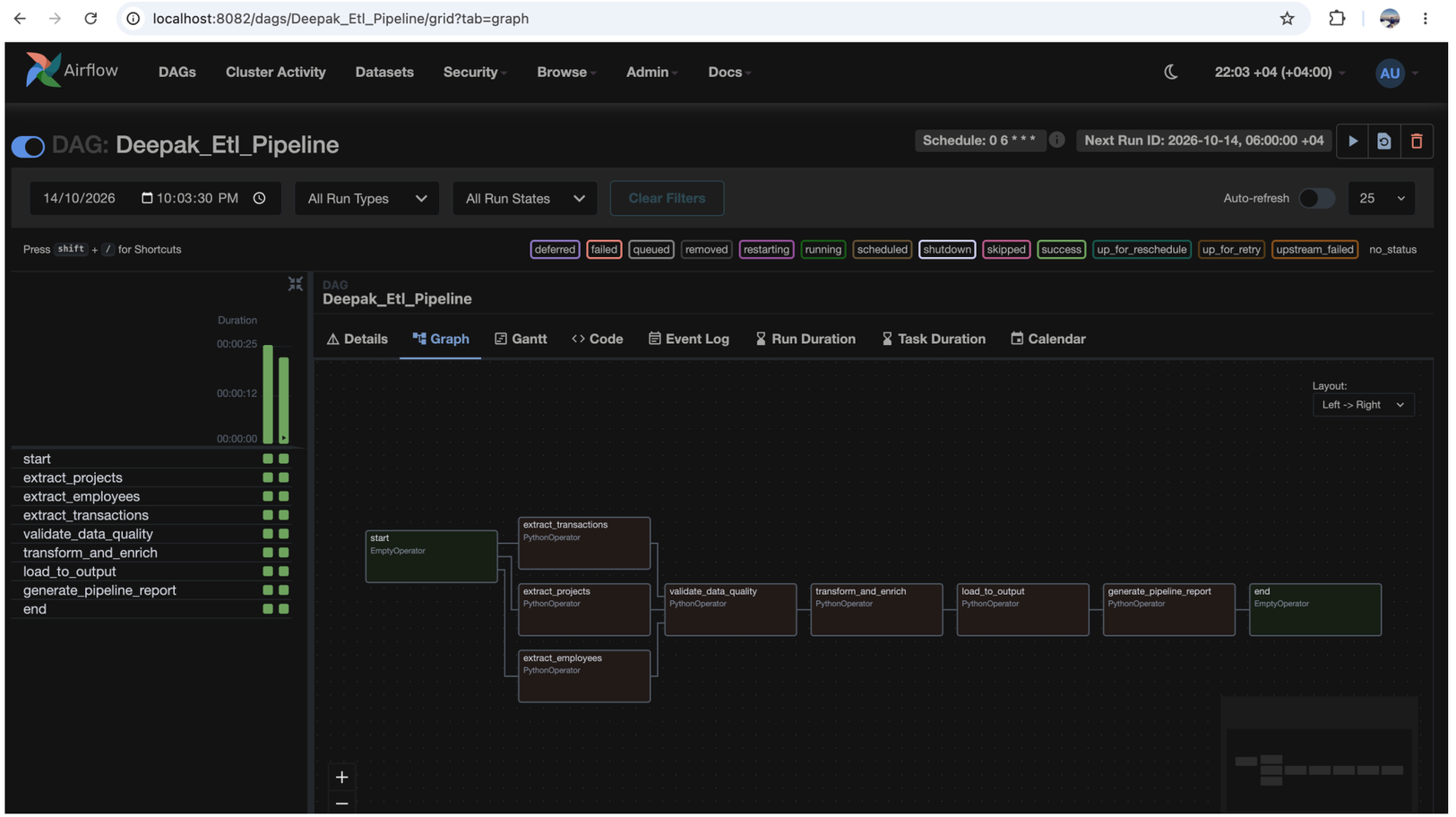


Dashboard preview

PILLAR 03

Big Data Processing

Task 3.3 | Airflow schedules the ETL and controls the order of work.



Green status cells show two successful runs. Schedule: daily at 06:00 Dubai time.

3.1 Spark: five Parquet summaries from 99,996 events across 12 monthly files.

3.2 Kafka: 8,333 January events consumed; 14 Critical escalations forwarded; JSON summary saved.

WHAT THE WORKFLOW DOES

1. Parallel extraction

Projects, employees and transactions are read independently.

2. Quality gate

Checks keys and references; completeness below 80% fails the gate.

3. Transform and enrich

Cleans source data and adds project / approver context.

4. Load outputs

Writes cleaned CSV files to the shared output folder.

5. Generate report

Uses XCom counts, quality results and paths to write the run report.

PILLAR 04

Infrastructure & Governance

Package the ETL, document data controls and make quality checks configurable.

TASK	QUESTION ASKED	WHAT I DID
4.1	Containerise ETL with Python 3.11, env paths and mounted outputs.	Created a multi-stage Dockerfile, .dockerignore and Compose override. Runs etl_starter.py, reads DATA_DIR/OUTPUT_DIR and exports host-accessible files; verified 50K transactions in 2.43 s.
4.2	Inventory/classify all four datasets; define governance controls.	Prepared data_governance_document.md: all 45 columns classified, PII/privacy scope documented, ownership, retention and access policies defined; lineage diagrammed.
4.3	Configurable quality rules, scores, warnings and reports.	Built rule_config.csv with six fields and 79 rules, plus a Python runner. Reports each rule's score in CSV/markdown/JSON; includes freshness and distribution. Nine tests passed.

GOVERNANCE DOCUMENT | SIX REQUIRED SECTIONS

Inventory • Classification • Ownership • Retention • Access control • Lineage
Salary history has restricted access and separate retention requirements.

Open governance document

Governance describes proposed policies; the local ETL does not enforce production access or retention.

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Thank you

Questions & discussion

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